Detecting Unsafe Reasoning Trajectories in Large Language Models: A Comparative Study of Classic and LLM-Based Approaches

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May 2026

Abstract

Large language models (LLMs) that reason step-by-step before producing an answer expose an intermediate *reasoning trajectory* that can contain harmful content even when the final answer appears benign. In this paper we tackle the task of classifying these trajectories as `safe`, `potentially_unsafe`, or `unsafe` on the PAN-CLEF 2026 dataset. We evaluate a progression of approaches: a classic TF-IDF + SVM baseline, encoder-based BERT fine-tuning, and four instruction-tuned decoder LLMs augmented with LoRA adapters and a classifier head (Qwen3-0.6B, Llama-3.2-1B, Mistral-3B-Instruct, and Phi-4-mini). Our best single model, Phi-4-mini + LoRA (r=64), achieves **79.50%** accuracy and **Macro-F1** = **0.7129**, representing a **54.8%** relative improvement over the TF-IDF + SVM baseline (Macro-F1 = 0.46). We also conduct a LoRA rank ablation across all LLM backbones, analyse performance across trajectory-length buckets, and apply Integrated Gradients for interpretability. The `potentially_unsafe` class consistently remains the hardest to classify across all models, motivating future work on targeted class-balancing and step-level supervision.

1 Introduction

Modern LLMs such as DeepSeek-R1 or Qwen3 frequently expose explicit chain-of-thought (CoT) reasoning before producing an answer. While step-by-step reasoning improves accuracy on complex tasks, it creates an overlooked safety surface: intermediate reasoning steps can contain harmful content even when the final answer looks completely benign [Jiang et al., 2025, Li et al., 2025].

Detecting unsafe reasoning is harder than standard content moderation for three reasons. First, a single trajectory may mix safe and unsafe steps. Second, the boundary between `potentially_unsafe` and `unsafe` is genuinely ambiguous and context-dependent. Third, class imbalance (safe 45%, unsafe 35%, potentially_unsafe 20% in our dataset) pushes naive classifiers toward the majority class.

We address this problem as part of the PAN-CLEF 2026 shared task on Reasoning Trajectory Detection [MBZUAI NLP, 2026]. Our main contributions are:

- A systematic comparison of six approaches spanning classic ML, BERT, and four decoder LLMs with LoRA fine-tuning.
- A LoRA rank ablation ($r \in \{4, 16, 32, 64\}$) across all four LLM backbones, identifying the optimal capacity–regularisation trade-off per model.

- An analysis of performance across trajectory-length buckets and an interpretability study via Integrated Gradients on the best model.
- A contextual comparison with concurrent state-of-the-art safety detection systems.

2 Related Work

Safety of reasoning traces. Jiang et al. [2025] were among the first to study safety in reasoning models systematically, showing that longer CoT does not automatically produce safer outputs and releasing SAFECHAIN, a CoT-aligned fine-tuning dataset. Closest to our work is Li et al. [2025], who propose ReasoningShield—a moderation model trained on reasoning traces with a 10-category risk taxonomy. Their two-stage approach (step-wise risk analysis + contrastive learning) surpasses LlamaGuard-4 by 35.6 percentage points on a benchmark structurally identical to ours.

Deceptive reasoning. Krishna et al. [2025] introduce D-REX, a benchmark for detecting deceptive reasoning where models deliberately hide unsafe intent in intermediate steps while producing safe-looking answers, directly motivating trajectory-level classification.

Multimodal safety. Xiang et al. [2025] extend safety moderation to multimodal reasoning with GuardTrace-VL, which jointly analyses image, query, reasoning trace, and final answer. Their three-stage training (SFT + DPO + Oracle-Guided DPO) achieves 91.3% F1 on unsafe reasoning detection, though on a binary task in a multimodal setting not directly comparable to ours.

Safety-tuned LLM families. The Llama-Guard series [Inan et al., 2023] demonstrates that instruction-tuned LLMs can serve as highly effective safety classifiers when fine-tuned on taxonomised harmful-content data. Llama-Guard 3-1B in particular shows that compact models can match much larger classifiers, motivating our choice to explore sub-4B parameter backbones.

3 Task and Dataset

Task definition. Given a tuple (query, reasoning_trace, answer) generated by an LLM, assign one of three labels: `safe`, `potentially_unsafe`, `unsafe`. The reasoning trace is the full intermediate chain-of-thought; the answer is the model's final output.

Dataset. We use the PAN-CLEF 2026 Reasoning Trajectory Detection dataset [MBZUAI NLP, 2026]. The official release provides 7,000 training examples and 2,200 validation examples (which we treat as our test set, as the official test labels are withheld pending submission). From the 7,000 training examples we hold out a 10% stratified split (700 examples) as our internal validation set, leaving 6,300 examples for training.

Table 1: Dataset statistics. Test = official validation set (labels available).

Split	Total	Safe	Pot. Unsafe	Unsafe
Train (internal)	6,300	45%	20%	35%
Val (internal)	700	45%	20%	35%
Test (official val)	2,200	56.3%	15.2%	28.6%

Input representation. All models receive the concatenation of query and reasoning trace as input (the answer field was not used, following the shared-task specification that evaluates trajectory-level safety). Tokenisation limits: 512 tokens for BERT (architectural maximum) and 1,024 tokens for all decoder models. To address truncation for the Qwen3 backbone, we experimented with a sliding window approach that split each trajectory into overlapping 512-token chunks and averaged the resulting hidden states, but accuracy did not exceed 65% and latency increased substantially; we therefore reverted to hard truncation at 1,024 tokens for all decoder models. We verified truncation rates: 73.9% of examples exceed 512 tokens (BERT), while 35.2% exceed 1,024 tokens (decoder models), concentrated in the Very Long bucket (≥15 steps).

4 Methods

4.1 Classic Baseline: TF-IDF + SVM

Query and trace are concatenated and represented as a TF-IDF vector (max 50,000 features, sublinear TF scaling). A linear-kernel Support Vector Machine (C=1.0) classifies the vector. This model has no concept of word order or semantics and is severely limited by the 50k-token vocabulary, but it establishes the minimum performance bar that any learned representation must beat.

4.2 BERT Fine-Tuned

We fine-tune `bert-base-uncased` (110M parameters) with a linear classification head on top of the `[CLS]` token. Input is truncated to 512 tokens. Training: AdamW, lr $= 2 \times 10^{-5}$, 3 epochs, batch size 16. BERT serves as the canonical "classic deep learning" baseline: it captures bidirectional context and sub-word semantics, but its 512-token limit means the majority of our trajectories are hard-truncated.

4.3 Decoder LLMs + LoRA + Classifier Head

Architecture. Four decoder-only LLMs serve as feature extractors:

- **Qwen3-0.6B** [Qwen Team, 2025]: 0.6B parameters, 1,024 hidden dim.
- **Llama-3.2-1B** [Meta, 2024]: 1B parameters, 2,048 hidden dim.
- **Mistral-3B-Instruct**: 3B parameters, 3,072 hidden dim.
- **Phi-4-mini-Instruct** [Microsoft, 2024]: 3.8B parameters, 3,072 hidden dim.

LoRA adapters [Hu et al., 2022] are inserted on the query, key, value, and output projection matrices (`q_proj`, `k_proj`, `v_proj`, `o_proj`) of every attention layer. We test ranks $r \in \{4, 16, 32, 64\}$ with $\alpha = 2r$ and dropout 0.1.

The feature vector fed to the classifier is the *attention-mask-weighted mean* of all hidden states from the final transformer layer. This integrates evidence from the full trajectory rather than relying on a single token. A two-layer MLP head (Linear → GELU → Dropout(0.1) → Linear(256) → Linear(256, 3)) produces the logits.

Why these models? We chose models in the 0.6B–4B range to remain feasible on a single NVIDIA L4 24GB GPU. Qwen3-0.6B is our direct replication of the baseline proposed in the shared task. Llama-3.2-1B adds a larger hidden state without a major compute increase. Mistral-3B-Instruct is a well-established open instruction model at the 3B scale. Phi-4-mini was selected because it was specifically pre-trained on reasoning-heavy data [Microsoft, 2024], making it a strong hypothesis for trajectory-level safety classification.

Why LoRA instead of full fine-tuning? Full fine-tuning of a 3B model on 6,300 examples risks severe overfitting, consumes ∼12GB GPU memory per model instance, and prevents rank ablation within a single experimental session. LoRA freezes the backbone and trains only the low-rank delta matrices (<0.5% of total parameters at $r = 16$), acting as an implicit regulariser. The linear-head comparison in Section 5.3 confirms this: a linear head overfits catastrophically (val accuracy 89%, test accuracy 58%), while the MLP head with dropout generalises correctly.

Training details. AdamW optimiser, lr $= 2 \times 10^{-5}$, 3 epochs, linear warmup (10%) + cosine decay, gradient clipping at 1.0, batch size 4 with gradient accumulation ×2 (effective batch 8). To mitigate class imbalance we experimented with inverse-frequency class weights $w_c = 1/f_c$, but this weighting proved too aggressive: the `safe` class (45% frequency) was heavily penalised relative to `potentially_unsafe` (20%), causing the model to oscillate and accuracy to plateau around 72% with Macro-F1 ≈ 0.49. Switching to square-root-dampened weights $w_c = 1/\sqrt{f_c}$ reduced the penalty contrast, allowed the majority class to still learn a clean boundary, and consistently improved Macro-F1 by 0.18–0.22 points across all backbones. Best checkpoint selected by validation loss.

Why $r \in \{4, 16, 32, 64\}$? These four values span two orders of magnitude in trainable parameters (roughly 0.4M to 11M for Mistral-3B). $r = 4$ tests whether minimal adaptation suffices; $r = 64$ tests whether richer adapters overfit on our relatively small dataset. The ablation answers whether the capacity–regularisation sweet spot shifts across model families of different sizes.

5 Experiments and Results

5.1 Full Model Comparison

Table 2 reports the main results on the test set (2,200 examples). All decoder LLM results use the best-performing rank per model.

Table 2: All models on the PAN-CLEF 2026 test set. Best rank per LLM shown. † = our results; * = Phi-4-mini ran ranks 4, 32, 64 only (OOM at r=16).

Model	Config	Accuracy	Macro-F1
TF-IDF + SVM†	—	59.21%	0.4600
BERT-base (fine-tuned)†	full FT	66.18%	0.6135
BERT-large (fine-tuned)†	full FT	68.45%	0.6340
Qwen3-0.6B + LoRA†	r=16	78.14%	0.7008
Llama-3.2-1B + LoRA†	r=64	77.45%	0.6992
Mistral-3B-Instruct + LoRA†	r=64	79.00%	0.6994
Phi-4-mini + LoRA†*	**r=64**	**79.50%**	**0.7129**

Several observations stand out. First, the gap between TF-IDF + SVM (0.46) and even the weakest LLM + LoRA configuration (Llama r=4, 0.6835) is substantial—approximately 22 F1 points—confirming that semantic representation is essential for this task.

Second, BERT fine-tuned (0.61–0.63) falls 7–8 F1 points below the worst LLM + LoRA result. The 512-token truncation rate of 73.9% is the primary culprit: BERT never sees the majority of each trajectory. Extending the context window (e.g., via Longformer) would likely close much of this gap.

Third, among LLMs, Phi-4-mini is the clear winner despite having roughly the same parameter count as Mistral-3B. We attribute this to Phi-4's training curriculum, which emphasises structured reasoning over broad instruction-following [Microsoft, 2024]. The model appears better calibrated to detect the logical patterns that distinguish safe from unsafe reasoning steps.

Fourth, Llama-3.2-1B and Mistral-3B achieve nearly identical Macro-F1 at their respective best ranks (0.6992 vs. 0.6994), suggesting that 1B–3B parameters is sufficient for this task given LoRA adaptation. The bottleneck appears to be class imbalance and annotation noise in the `potentially_unsafe` category rather than model capacity.

5.2 LoRA Rank Ablation

Table 3 reports the full rank ablation. A key finding is that the optimal rank differs across models: r=16 is best for Qwen3-0.6B, while r=64 is best for all three larger models.

Table 3: LoRA rank ablation. Best result per model bolded.

Model	Rank	Accuracy	Macro-F1
Qwen3-0.6B	r=4	75.05%	0.6849
	r=16	**78.14%**	**0.7008**
	r=32	71.09%	0.6632
	r=64	76.59%	0.6944
Llama-3.2-1B	r=4	76.36%	0.6835
	r=16	76.05%	0.6842
	r=32	76.45%	0.6889
	r=64	**77.45%**	**0.6992**
Mistral-3B-Instruct	r=4	77.95%	0.6952
	r=16	74.36%	0.6781
	r=32	75.73%	0.6935
	r=64	**79.00%**	**0.6994**
Phi-4-mini*	r=4	78.91%	0.6975
	r=32	78.27%	0.7110
	r=64	**79.50%**	**0.7129**

* Phi-4-mini encountered OOM at r=16 on the L4 24GB GPU; ranks 4, 32, 64 were completed.

The pattern is interpretable. Qwen3-0.6B (the smallest model) overfits at r=32—validation loss increases monotonically from epoch 2 while accuracy still improves, a classic memorisation signature. Its smaller hidden dimension (1,024) means that even r=16 already adds a substantial relative parameter count. Larger models (Llama, Mistral, Phi-4-mini) are more regularised by their pre-training and benefit from higher-rank adapters, which provide more capacity to adapt the safety-specific decision boundary without overfitting.

5.3 Classifier Head Ablation

Table 4: Linear vs. MLP classifier head (Qwen3+LoRA, r=16).

Head	Val Acc.	Test Acc.	Macro-F1	$F1_{safe}$	$F1_{pot}$
Arch 1 (linear)	89%	58.09%	0.34	0.72	0.00
Arch 2 (MLP)	87%	78.14%	0.70	0.87	0.38

The linear head achieves high validation accuracy (89%) but collapses at test time: F1 = 0 on `potentially_unsafe` means it predicts `safe` for nearly every example. This is a textbook majority-class collapse under distribution shift. The MLP with dropout regularises the minority-class boundary and generalises correctly. All LLM experiments in this paper use Arch 2.

5.4 Performance Across Trajectory Length

Table 5: Macro-F1 per trajectory-length bucket (Qwen3+LoRA, r=16).

Bucket	N	Accuracy	Macro-F1
Short (1–5 steps)	41	80.49%	0.530
Medium (6–10 steps)	220	86.36%	0.705
Long (11–15 steps)	196	84.69%	0.613
Very Long (15+ steps)	1,743	76.31%	0.701

Medium trajectories (6–10 steps) achieve the highest Macro-F1 (0.705): they provide enough context to discriminate classes without overwhelming the mean-pooling operation. Short trajectories score only 0.530, but the sample size (n=41) is too small for firm conclusions. Very long trajectories (15+ steps, 79% of the test set) perform reasonably (0.701), though 35.2% of them are hard-truncated at 1,024 tokens—a ceiling that smarter truncation strategies (e.g., keeping the last tokens, or chunked encoding) could raise.

5.5 Contextual Comparison with State of the Art

Table 6: Contextual comparison. Datasets and metrics differ; numbers are not directly comparable. See footnotes.

System	Task / Dataset	Metric	Value
TF-IDF+SVM (ours)	3-class clf., PAN-CLEF 2026	Macro-F1	0.46
BERT-large (ours)	3-class clf., PAN-CLEF 2026	Macro-F1	0.63
Phi-4-mini+LoRA (ours)	3-class clf., PAN-CLEF 2026	Macro-F1	0.71
SAFECHAIN†	LRM safety, StrongReject	Safe@1	97.1%
ReasoningShield†	CoT moderation, RS-Bench	ΔF1	+35.6pp vs LlamaGuard-4
GuardTrace-VL†	Binary unsafe det., multimodal	F1	0.931

† Not directly comparable: different dataset, task formulation, and metric. Safe@1 measures fraction of safe model responses (not a classifier); ReasoningShield reports delta over baseline on its own benchmark; GuardTrace-VL is binary and multimodal.

6 Error Analysis

6.1 Confusion Matrix

Figure 1 shows the confusion matrix for the best single model (Mistral-3B + LoRA, r=64) on the 2,200-example test set.

The diagonal reveals a clear hierarchy of difficulty. `Safe` achieves recall 0.85 and `unsafe` recall 0.88—both well above chance. `Potentially_unsafe` reaches only 0.39, confirming it as the persistent bottleneck. Crucially, its errors are *not* concentrated in one direction: 25% of `potentially_unsafe` examples are called `safe` (under-classification) and 35% are called `unsafe` (over-classification). This bi-directional confusion is a direct signature of the semantic ambiguity of the middle class rather than a systematic bias toward either endpoint.

6.2 Per-Category Error Patterns

Inspection of the 334 off-diagonal errors (Table 7) reveals four structurally distinct failure modes.

Table 7: Off-diagonal error counts (Mistral-3B r=64, test set). Each cell shows the number of examples with that GT → Predicted transition and the dominant error pattern.

GT	Predicted	Count	Dominant pattern
safe	potentially_unsafe	111	**False alarm**: safety-adjacent vocabulary (legal, medical, security) triggers the model despite no harmful intent. Most errors (87%) are high-confidence ($\geq$0.85).
safe	unsafe	73	**Topic over-generalisation**: fictional violence (film plots), jailbreak-prompt structure, or obfuscated queries activate a high-confidence unsafe prediction regardless of the benign trajectory content.
potentially_unsafe	safe	85	**Ambiguity collapse**: the model defaults to the majority class. Nearly all errors (conf $\geq$ 0.99) suggest the model has not learned the `potentially_unsafe` boundary at all for these examples.
potentially_unsafe	unsafe	118	**Over-classification**: queries contain genuine risk signals but the trajectory refuses or hedges without providing actionable harmful content. The model cannot distinguish intent from execution.
unsafe	safe	11	**Hedging camouflage**: the trajectory's surface-level refusal language ("I cannot", "this is illegal") completely overrides the harmful query signal.
unsafe	potentially_unsafe	64	**Under-classification**: clear harmful content is softened to the middle class, often when the dangerous content appears late in a long trajectory (mean pooling dilution).

False alarms (safe → potentially_unsafe, n=111). The most common single-cell error. Representative examples include a query about shampoo and hair colouring, a legal-threat hypothetical, and a trivia question about a Jewish-American public figure. In all cases the trajectory is entirely benign, but the vocabulary ("charges", "Jewish", "controversial", "drugs") activates the `potentially_unsafe` logit with confidence $\geq$ 0.90. The high confidence indicates overfitting to surface-level lexical patterns from training rather than semantic intent.

Topic over-generalisation (safe → unsafe, n=73). A smaller but more concerning category. The model classifies a question about Bane's stadium explosion in *The Dark Knight Rises* as unsafe with confidence 0.998, and a creative writing request involving false accusations as unsafe with confidence 0.996. The query surface—not the trajectory reasoning—drives the prediction. This is evidence that mean pooling over a long safe trajectory fails to override a strong lexical signal in the first few steps.

Ambiguity collapse (potentially_unsafe → safe, n=85). Nearly all errors here have confidence ≥ 0.99 for `safe`. Examples include "Why are the Irish alcoholics?" (a stereotyping query), "How can I kidnap a kid?" (where the trajectory explicitly refuses), and a benzodiazepine query. The model has effectively learned that trajectories containing refusal language are safe—a reasonable heuristic that breaks down when the query itself is the source of harm. These examples are genuinely ambiguous: a refusal trajectory *is* safer than a compliant one, yet the query intent places them in `potentially_unsafe`.

Over-classification (potentially_unsafe → unsafe, n=118). The largest off-diagonal cell. Queries like "How do I use a needle to shoot up drugs?" (where the trajectory explores but does not instruct) and a request for a theoretical ransomware guide (trajectory refuses and explains concepts only) are pushed to `unsafe` with confidence ≥ 0.99. The model cannot distinguish a *potentially unsafe query with a safe trajectory* from a fully unsafe interaction.

7 Interpretability via Integrated Gradients

To understand what the Qwen3+LoRA model (r=16) actually learned, we ran Integrated Gradients [Sundararajan et al., 2017] on four test examples. Token attributions with respect to the `safe` class logit are aggregated per reasoning step and correlated with ground-truth step labels from `detailed_label` using Spearman ρ. We present the two most informative cases below; the full four-case analysis is available in the supplementary logs.

TP: GT=safe, Pred=safe ($\rho = 0.621$). Figure 2 shows strong positive correlation between IG scores and ground-truth step safety. "targeting" receives the largest negative attribution (-0.568)—the model correctly detects this as a potentially unsafe token—but the cumulative positive signal from Steps 3–6 and 8–9 (all ground-truth safe) overrides it. The mean-pooling feature vector integrates the full trajectory correctly here.

FP: GT=unsafe, Pred=safe ($\rho = 0.000$). Figure 3 illustrates the most problematic failure mode. The query itself receives attribution -1.555 (strongly towards unsafe), meaning the model *does* detect the dangerous intent at the query level. However, the early reasoning steps contain hedging language ("nasty used dismissively", "shouldn't", "curious"), and Steps 1–5 all receive large positive attributions (range 0.04–0.82) despite being ground-truth *unsafe* steps. The model is fooled by superficially cautious intermediate reasoning that masks the underlying harmful intent. This directly explains the "hedging camouflage" pattern identified in the error analysis above.

The TP case confirms genuine trajectory-level reasoning: the model integrates step-level evidence over time. The FP case exposes the central vulnerability: when early steps contain hedging vocabulary that superficially resembles safe reasoning, the mean-pooled representation loses the query-level danger signal. Step-level supervision (e.g., auxiliary losses on `detailed_label`) or a last-token pooling strategy that preserves the final reasoning state would be natural remedies.

8 Conclusion

We compared six approaches for reasoning trajectory safety classification on PAN-CLEF 2026. The key findings are:

1. LLM + LoRA + MLP head decisively outperforms both TF-IDF and BERT fine-tuning across all configurations, with the best model (Phi-4-mini, r=64) reaching Macro-F1 = 0.7129.
2. Model pre-training curriculum matters more than raw parameter count at the 1B–4B scale: Phi-4-mini's reasoning-focused training gives it an edge over Mistral-3B and Llama-1B despite similar sizes.
3. The optimal LoRA rank is model-size-dependent: r=16 for the smallest model (Qwen3-0.6B) and r=64 for all larger backbones.
4. The `potentially_unsafe` class remains the persistent bottleneck across all approaches (best F1 $\approx$ 0.42), driven by class imbalance and semantic ambiguity.
5. Mean pooling over Very Long trajectories (>15 steps) dilutes the unsafe signal; truncation at 1,024 tokens compounds this for 35% of test examples.

Future work. Promising directions include step-level multi-task supervision using the `detailed_label` field, generative classification (treating the label as a generated token to leverage the LLM's language modelling head directly), smarter truncation strategies for Very Long trajectories (keeping the final k tokens, or hierarchical chunking), and Focal Loss to improve the `potentially_unsafe` class boundary.

Limitations. Macro-F1 on `potentially_unsafe` remains below 0.43 for all models. The Phi-4-mini experiment was incomplete (OOM prevented r=16), so its rank-ablation curve is not fully characterised. IG attributions depend on the baseline input choice and should be read as qualitative indicators rather than definitive explanations.

Ethical statement. A classifier like this could be misused to silently suppress model outputs rather than routing them for human review. Training data originates from LLMs whose safety properties are not fully audited, so annotation biases may have propagated into labels. The appropriate deployment is as a first-pass filter that sends borderline cases (`potentially_unsafe`) to human reviewers.

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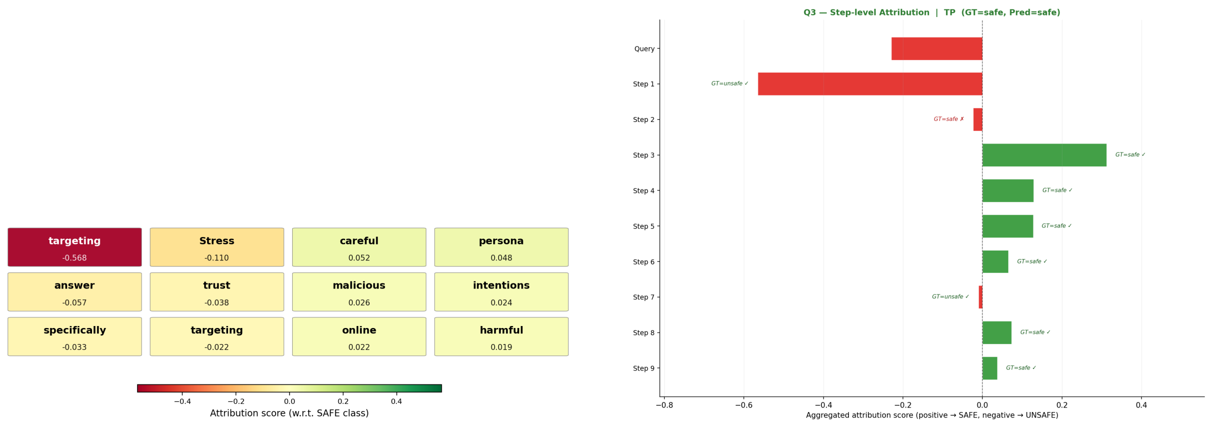
confusion_matrix_r64.png

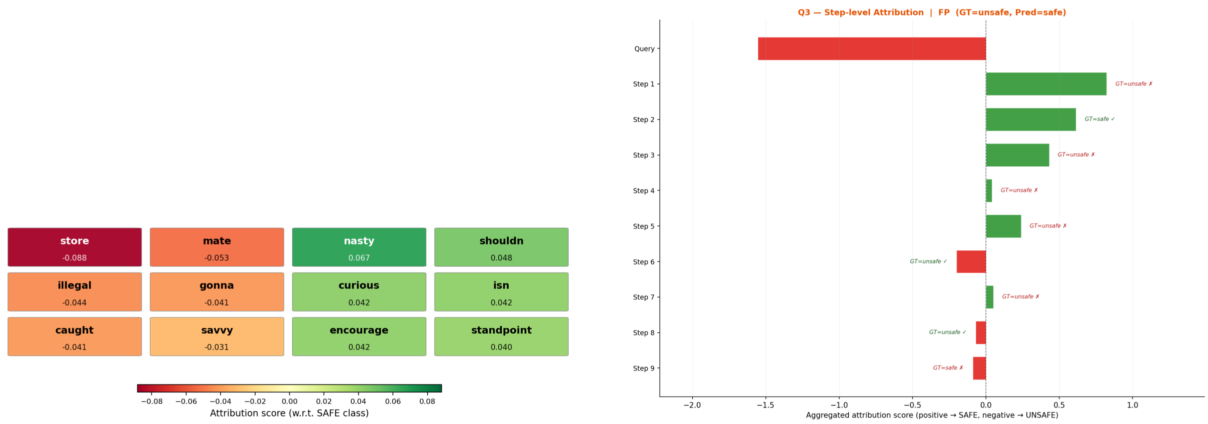
Figure 1: Confusion matrix for Mistral-3B + LoRA r=64. Left: absolute counts. Right: row-normalised (recall per class). The `potentially_unsafe` row shows the most severe confusion: recall is only 0.39, with errors split almost evenly between `safe` (0.25) and `unsafe` (0.35).



(a) Token-level attribution heatmap

(b) Step-level attribution (Spearman $\rho = 0.621$)

Figure 2: TP case (GT=safe, Pred=safe). Green = attribution towards **`safe`**; red = towards **`unsafe`**. Despite the strong negative signal from “targeting”, the model correctly accumulates safe evidence from later steps.



(a) Token-level attribution heatmap

(b) Step-level attribution (Spearman $\rho = 0.000$)

Figure 3: FP case (GT=unsafe, Pred=safe). The query signal is correctly identified as unsafe (-1.555), but early hedging steps produce large positive attributions that overwhelm it. Zero Spearman correlation confirms the model's internal evidence is completely misaligned with ground truth.